

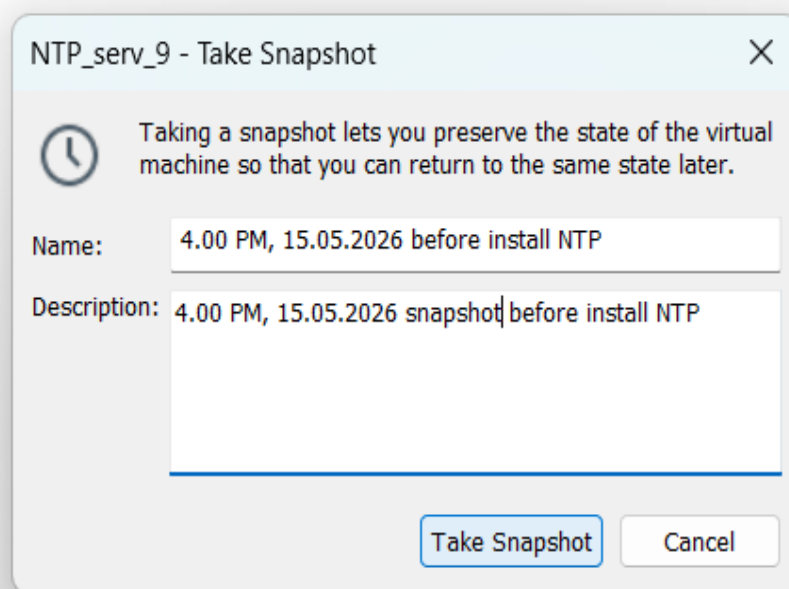
Aim :-

To install, configure and test NTP (Network Time Protocol) server . it is used to sync system time.

Step 1: log in (server site)

Step 2: Take Backup and Restore Plan.

Snapshot



Step 3 : check Network and configure it

```

[root@ntpserver ~]# hostname
ntpserver
[root@ntpserver ~]# nmtui
[root@ntpserver ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.100 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::20c:29ff:feab:6e48 prefixlen 64 scopeid 0x20<link>
    inet6 2409:40c2:123c:36ea:20c:29ff:feab:6e48 prefixlen 64 scopeid 0x0<global>
    ether 00:0c:29:ab:6e:48 txqueuelen 1000 (Ethernet)
    RX packets 1758 bytes 461982 (451.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 606 bytes 61254 (59.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 37 bytes 3846 (3.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 37 bytes 3846 (3.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@ntpserver ~]#
[root@ntpserver ~]#
[root@ntpserver ~]# █

```

Step 4 : configure required Repository

```

[root@ntpserver yum.repos.d]# yum install -y chrony
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

this is BaseOS repo
this is AppStream repo
Last metadata expiration check: 0:00:01 ago on Thu 14 May 2026 05:17:56 PM IST.
Package chrony-4.6.1-2.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ntpserver yum.repos.d]#
[root@ntpserver yum.repos.d]#
[root@ntpserver yum.repos.d]# █

```

	37 MB/s		2.2 MB	00:00
	53 MB/s		7.9 MB	00:00

Step 5 : Edit configuration file

#vi /etc/chrony.conf

Make change : allow 192.168.0.0/24

```
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (https://www.pool.ntp.org/join.html).
#pool 2.rhel.pool.ntp.org iburst
allow 192.168.0.0/24 iburst

# Use NTP servers from DHCP.
sourcedir /run/chrony-dhcp

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync
```

Step 6 : Start and Enable service and check status and verify Time Sync .

```
root@ntpserver yum.repos.d]# systemctl start chronyd
root@ntpserver yum.repos.d]# systemctl enable chronyd
root@ntpserver yum.repos.d]# systemctl status chronyd
chronyd.service - NTP client/server
Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: enabled)
Active: active (running) since Thu 2026-05-14 16:37:18 IST; 53min ago
Docs: man:chronyd(8)
      man:chrony.conf(5)
Main PID: 810 (chronyd)
Tasks: 1 (limit: 10312)
Memory: 4.2M (peak: 4.8M)
CPU: 184ms
CGroup: /system.slice/chronyd.service
        └─810 /usr/sbin/chronyd -F 2

lay 14 16:41:08 ntpserver chronyd[810]: Source 2600:3c16::2000:a0ff:fe85:2df0 offline
lay 14 16:41:08 ntpserver chronyd[810]: Source 2605:e440:19::1e7 offline
lay 14 16:41:08 ntpserver chronyd[810]: Source 192.46.210.39 offline
lay 14 16:41:08 ntpserver chronyd[810]: Can't synchronise: no selectable sources (4 unreachable sources)
lay 14 16:41:08 ntpserver chronyd[810]: Source 2001:678:8::123 offline
lay 14 16:41:08 ntpserver chronyd[810]: Source 192.46.210.39 online
lay 14 16:41:08 ntpserver chronyd[810]: Source 192.46.210.39 offline
lay 14 16:41:09 ntpserver chronyd[810]: Source 192.46.210.39 online
lay 14 16:41:27 ntpserver chronyd[810]: Source 192.46.210.39 replaced with 2605:e440:19::342 (2.rhel.pool.ntp.org)
lay 14 16:47:51 ntpserver chronyd[810]: Selected source 2605:e440:19::342 (2.rhel.pool.ntp.org)
root@ntpserver yum.repos.d]# c
ash: c: command not found...
root@ntpserver yum.repos.d]#
root@ntpserver yum.repos.d]# chronyc sources
IS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
? 2605:e440:19::1e7          2    6      0   51m  -1942us[ +13ms] +/- 126ms
* 2605:e440:19::342          2    8    377   118  -3698us[-4656us] +/- 122ms
? any.time.nl                2    6      0   51m  +2001us[+8539us] +/-  55ms
? 2600:3c16::2000:a0ff:fe8>  4    6      0   51m  -464us[+6003us]  +/-  54ms
root@ntpserver yum.repos.d]#
```

Temporarily disable firewall

```
[root@ntpserver yum.repos.d]# systemctl stop firewalld.service
[root@ntpserver yum.repos.d]# setenforce 0
[root@ntpserver yum.repos.d]# getenforce
Permissive
[root@ntpserver yum.repos.d]# █
```

Step 7 :Remote testing from different Machine :

1) On Client (rhel 8) machine install Package :-

```
[root@ntpserver yum.repos.d]# yum install -y chrony
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

this is BaseOS repo
this is AppStream repo
Last metadata expiration check: 0:00:01 ago on Thu 14 May 2026 05:17:56 PM IST.
Package chrony-4.6.1-2.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ntpserver yum.repos.d]#
[root@ntpserver yum.repos.d]#
[root@ntpserver yum.repos.d]# █
```

37 MB/s	2.2 MB	00:00
53 MB/s	7.9 MB	00:00

2) Edit configure file:-

```
#vi /etc/chrony.conf (add : server 192.168.0.100
iburst)
```

```
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (http://www.pool.ntp.org/join.html).
#pool 2.rhel.pool.ntp.org iburst
server 192.168.0.100 iburst

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync

# Enable hardware timestamping on all interfaces that support it.
#hwtimestamp *

# Increase the minimum number of selectable sources required to adjust
# the system clock.
#minsources 2

# Allow NTP client access from local network.
#allow 192.168.0.0/16
```

3) Restart Service and Verify Client Sync :-

```
[root@nfs_server yum.repos.d]# systemctl restart chronyd
[root@nfs_server yum.repos.d]# systemctl enable chronyd
[root@nfs_server yum.repos.d]# chronyc sources
VS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^? 192.168.0.100             0    7    0    -    +0ns[  +0ns] +/-    0ns
[root@nfs_server yum.repos.d]#
```